

Laboratory Report

EIT & MEC – Digi Lab 2: Sequential Circuits

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Abstract

[Write after completing all experiments – max. 200 words. Cover: objective, circuits built, key results (e.g. monoflop timing, counter behaviour), main conclusion.]

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1 Introduction

Sequential circuits, unlike purely combinational circuits, possess memory: their output depends not only on the current inputs but also on the history of previous inputs stored in flip-flops. They are the basis of all digital systems that store state, count events, or synchronise operations to a clock signal.

This laboratory exercise investigates three fundamental sequential building blocks: the **shift register**, the **monoflop**, and the **synchronous counter**. A shift register stores a sequence of bits and moves them one position on each rising clock edge. A monoflop (one-shot) generates a pulse of defined duration in response to a trigger event. A synchronous BCD counter increments its binary output by one on each rising clock edge, resetting to zero after reaching nine.

All three components are combined in a progressive design task: first a chaser (moving) light is implemented, then extended with a catch function, and finally augmented with a catch counter displayed in BCD code.

The objectives of this laboratory are:

- To understand the operating principle of shift registers, monoflops, and synchronous counters.
- To design, simulate in LTSpice, and assemble three sequential circuits on a breadboard.
- To implement a power-on reset circuit ensuring a defined initial state after power-up.
- To debounce a push-button using a D-type flip-flop and gate logic.
- To combine multiple ICs into a functional system and verify correct end-to-end behaviour on the breadboard.

The theoretical background was reviewed according to [1] and [2].

2 Equipment and Materials

All circuits were assembled on a standard experimentation (breadboard) system. A square-wave clock signal of approximately 1 Hz, 5 V_{pp}, and 2.5 V DC offset was supplied by a waveform generator. A Schmitt-trigger inverter (74HC132) was inserted between the waveform generator and the logic circuit to ensure correct logic levels. Table 1 lists the equipment and ICs used.

Table 1: Equipment and components used in this experiment.

Item	Specification / Part No.
Digital oscilloscope	Teledyne LeCroy HD 4096
Oscilloscope probe (10:1)	—
Waveform generator	[Model]
DC power supply	5 V
Experimentation board	—
74HC00 (4x 2-in NAND)	74HC00
74HC74 (2x D-type flip-flop)	74HC74
74HC132 (4x NAND Schmitt)	74HC132
74HC160 (4-bit decade counter)	74HC160
74HC164 (8-bit shift register)	74HC164
74HC221 (2x monoflop)	74HC221
LEDs (8x chaser + 4x BCD)	—
Push-button	—
Resistors, capacitors	see individual tasks

3 Theoretical Background

3.1 Shift Register (74HC164)

A shift register is a chain of D-type flip-flops sharing a common clock line. On each rising clock edge, the data bit present at the serial input is loaded into the first flip-flop, while every subsequent flip-flop takes over the value of its predecessor. The 74HC164 is an 8-bit serial-in/parallel-out shift register. Its data input is the AND combination of pins A and B; tying one permanently high makes the other the effective serial data input. An active-low asynchronous clear ($\overline{\text{CLR}}$) resets all flip-flops to zero immediately, regardless of the clock.

3.2 D-type Flip-Flop (74HC74)

The 74HC74 contains two independent positive-edge-triggered D-type flip-flops, each with active-low asynchronous preset ($\overline{\text{PR}}$) and clear ($\overline{\text{CLR}}$) inputs. On a rising clock edge, output Q takes the value present at input D . The asynchronous inputs override the clock: $\overline{\text{PR}}$ low forces $Q = 1$; $\overline{\text{CLR}}$ low forces $Q = 0$, independently of the clock or D .

3.3 Power-On Reset

After power-up, flip-flops settle in undefined states. A power-on reset (POR) circuit generates a short active reset pulse immediately after V_{CC} rises. An RC circuit ($R = 1 \text{ M}\Omega$, $C = 1 \text{ }\mu\text{F}$) charges slowly after power-on; the capacitor voltage is fed into a Schmitt-trigger inverter (74HC132). While V_C is below the Schmitt upper threshold U_S , the inverter output remains high (reset active, all flip-flops cleared). Once V_C exceeds U_S , the inverter output falls low (reset released). The switch-on delay is:

$$t_e = \tau = R \cdot C = 1 \text{ M}\Omega \times 1 \text{ }\mu\text{F} = 1 \text{ s}$$

This is long enough to allow the supply voltage to settle fully before the reset is released.

3.4 Monoflop (74HC221)

The 74HC221 is a dual retriggerable monoflop with active-low clear. It generates a precisely timed output pulse whose width is set by an external resistor R_{ext} and capacitor C_{ext} :

$$t_W = 0.7 \cdot R_{ext} \cdot C_{ext} \quad (1)$$

The monoflop is triggered by a rising edge at input B (with A low). During the pulse, $\overline{Q}$ is low (blocking the shift register clock); after the pulse, $\overline{Q}$ returns high (clock released).

3.5 BCD Decade Counter (74HC160)

The 74HC160 is a synchronous 4-bit decade counter that counts from 0 to 9 in natural binary (BCD), resetting to 0 on the next clock edge after 9. It features an active-low asynchronous clear ($\overline{CLR}$), a synchronous parallel load ($\overline{LOAD}$), and two count-enable inputs (ENP, ENT). The ripple-carry output (RCO) goes high at count 9 and can be used to cascade multiple counter stages.

4 Procedures

All tasks were carried out according to the laboratory instruction sheet *Lab_02_SequentialCircuits*, Rev. 1.0, and the design guide *Lab_02_SRC_Guide*, Rev. 1.1, both dated 05/06.05.2026.

For each task the following steps were followed:

1. Develop and draw the schematic based on the design guide.
2. Simulate the circuit in LTSpice using the provided model files (`74hc.lib`). Note: 74HC160 was replaced by 74HC161 in simulation only (identical pinout; counts to 15 instead of 9). For 74HC221 the SpiceLine parameter was changed from `TRIPDT=1e-9` to `TRIPDT=1e-1` to obtain a reasonable simulation time.
3. Present the schematic to the tutor for approval before assembly.
4. Assemble the circuit on the breadboard and verify functionality by observation.

The clock signal was set to 1 Hz, 5 V_{pp}, 2.5 V offset. All ICs were supplied with 5 V DC.

5 Results and Discussion

5.1 Task 3.1 – Chaser Light (8-bit Shift Register)

Component values used: $R_{POR} = 1\text{ M}\Omega$, $C_{POR} = 1\text{ }\mu\text{F}$, $V_{CC} = 5\text{ V}$, clock: 1 Hz.

5.1.1 Circuit Description and Design

The chaser light circuit uses the 74HC164 8-bit shift register as its core. One LED is connected via a current-limiting resistor to each of the eight parallel outputs (QA–QH). A single logic “1” is injected at startup and then shifted one position per clock cycle, illuminating exactly one LED at a time.

To inject exactly one “1” at startup and never again, the first flip-flop of the 74HC74 is configured as follows: the $\overline{CLR}$ input receives the active-low POR signal, the D input is

tied permanently to V_{CC} , and the clock input receives the same clock as the shift register. While the POR is active ($\overline{\text{CLR}}$ low), Q_{FF1} is held at zero. At the first rising clock edge after POR releases, Q_{FF1} goes high for exactly one clock period (since $D = \text{high}$), providing a single “1” to the shift register data input. Before the second rising edge, Q_{FF1} has returned low. This ensures that only one “1” is ever present in the shift register chain at any time.

To make the light wrap around after LED8, the serial output QH of the 74HC164 is fed back to the serial data input (pins A and B tied together).

The power-on reset RC circuit ($R = 1\text{ M}\Omega$, $C = 1\text{ }\mu\text{F}$, switch-on delay $t_e \approx 1\text{ s}$) drives a 74HC132 Schmitt-trigger inverter whose output provides the active-low $\overline{\text{CLR}}$ signal to both the 74HC74 and the 74HC164.

5.1.2 LTSpice Simulation

Figure 1: LTSpice simulation – chaser light: clock (top), POR signal, shift register data input, and outputs QA–QH.

The simulation confirms that exactly one output is high at any given time and that the “1” wraps correctly from QH back to QA. After power-on, all outputs remain low until the POR signal is released, after which QA is the first output to go high on the next rising clock edge.

5.1.3 Breadboard Verification

Figure 2: Assembled chaser light circuit on the breadboard.

[Describe the observed behaviour: did the single LED advance from LED1 to LED8 and wrap around correctly? Did power-on reset ensure LED1 was always first?]

5.2 Task 3.2 – Catch Circuit (Monoflop Extension)

Additional components: $R_{mono} = 2\text{ M}\Omega$, $C_{mono} = 1\text{ }\mu\text{F}$, $R_{pull} = 10\text{ k}\Omega$.

5.2.1 Circuit Description and Design

Task 3.2 extends the chaser light with a catch function: whenever LED5 is ON and the push-button is pressed, the moving light freezes at LED5 for approximately 2 s, after which it continues normally.

Three sub-circuits are added to the existing design.

Button Debounce and Catch Detection (74HC74, second flip-flop). The push-button is debounced and its effect gated by LED5 using the second D-type flip-flop of the 74HC74. The $\overline{\text{PR}}$ input is tied to V_{CC} . The $\overline{\text{CLR}}$ input is connected to the output of the LED5 shift register stage (Q5 of the 74HC164): when LED5 is OFF (Q5 low, hence

$\overline{\text{CLR}}$ low), Q_{FF2} is held at zero and no trigger can occur. The D input is tied to V_{CC} , and the clock input is connected to the push-button via a 10 k Ω pull-down resistor to GND. Only when LED5 is ON (Q_5 high, $\overline{\text{CLR}}$ high) can the flip-flop respond: on the first rising edge from the button, Q_{FF2} goes high and triggers the monoflop. This simultaneously debounces the button (only the first edge counts) and ensures the catch is only active when LED5 is lit.

Monoflop (74HC221) – Catch Timer. The 74HC221 monoflop generates the hold pulse. With $R_{ext} = 2 \text{ M}\Omega$ and $C_{ext} = 1 \text{ }\mu\text{F}$, the calculated hold time is:

$$t_{W,\text{calc}} = 0.7 \times 2 \text{ M}\Omega \times 1 \text{ }\mu\text{F} = 1.4 \text{ s} \quad (2)$$

$t_{W,\text{calc}} = 1.4 \text{ s}$

The monoflop is triggered by the rising edge of Q_{FF2} at input B (with A tied low, $\overline{\text{CLR}}$ connected to the POR signal). During the pulse, $\overline{Q}_{mono}$ is low.

Clock Blocking (74HC00 NAND Gate). A NAND gate from the 74HC00 blocks the clock signal to the 74HC164 during the monoflop pulse. One input of the NAND gate receives the waveform generator clock (via the 74HC132 Schmitt buffer); the other input receives $\overline{Q}_{mono}$ of the monoflop inverted (i.e. a second NAND gate used as an inverter, or directly Q_{mono} depending on the gate polarity chosen). While $\overline{Q}_{mono}$ is low, the NAND output is permanently high: no rising edges reach the shift register, and the light dot stays fixed at LED5 for the duration of the monoflop pulse.

5.2.2 LTSpice Simulation

Figure 3: LTSpice simulation – catch circuit: clock, Q_5 (LED5), button signal, Q_{FF2} , monoflop $\overline{Q}$, and blocked shift register clock.

The simulation confirms that the monoflop is triggered only when LED5 is ON and the button is pressed. The shift register clock is blocked for the duration of t_W , holding the light dot at position 5. Pressing the button while another LED is ON has no effect.

5.2.3 Breadboard Verification

Figure 4: Assembled catch circuit on the breadboard.

Table 2: Measured monoflop hold time vs. calculated value.

Quantity	Symbol	Calculated	Measured
Hold time	t_W	1.4 s	?? s

[Describe the observed behaviour: did LED5 stay on when the button was pressed during LED5? Did the light resume correctly afterwards? What happened when the button was pressed on a different LED?]

5.3 Task 3.3 – Catch Counter (BCD Counter Extension)

Additional components: 74HC160 decade counter, 4 LEDs with current-limiting resistors.

5.3.1 Circuit Description and Design

Task 3.3 extends the catch circuit by counting the number of successful catches and displaying the count in BCD code on 4 additional LEDs. The 74HC160 synchronous decade counter is used for this purpose.

Each time the light is caught, the monoflop generates a pulse. The rising edge of Q_{mono} (i.e. the end of the hold pulse, when the clock is released again) increments the counter by one. The four BCD outputs QA–QD of the 74HC160 drive 4 LEDs directly, displaying the count from 0 to 9 in binary.

The counter connections are:

- $\overline{\text{CLR}}$: connected to the POR signal (resets counter to 0 at power-on).
- ENP, ENT: both tied to V_{CC} (counting always enabled).
- $\overline{\text{LOAD}}$: tied to V_{CC} (parallel load disabled).
- CLK: connected to Q_{mono} (rising edge on each successful catch).
- QA–QD: connected to 4 LEDs via current-limiting resistors.

After 9 catches, the counter automatically resets to 0 on the next catch due to the decade counter's built-in modulo-10 behaviour.

5.3.2 LTSpice Simulation

Figure 5: LTSpice simulation – catch counter: monoflop Q output used as counter clock, and BCD outputs QA–QD of the 74HC160.

The simulation (using 74HC161 instead of 74HC160) confirms that the counter increments correctly on each monoflop rising edge. The BCD output was verified to advance from 0000 to 1001 (0–9) before wrapping back to 0000. The POR signal correctly initialises the counter to 0 at startup.

5.3.3 Breadboard Verification

Figure 6: Assembled catch counter circuit on the breadboard.

[Describe the observed behaviour: did the 4 LEDs display the correct BCD count after each catch? Did the counter reset to 0 after 9 catches? Did power-on reset initialise the counter to 0?]

6 Conclusions

The following table summarises the verification status of all three circuits:

Table 3: Summary of designed, simulated, and verified circuits.

Task	Circuit	t_W calc.	t_W meas.	Rel. error
3.1	Chaser light (74HC164 + 74HC74 + POR)	—	—	—
3.2	Catch function (+ 74HC221 + 74HC00)	1.4 s	?? s	?? %
3.3	Catch counter (+ 74HC160 + 4 LEDs)	—	—	—

[Write 4–6 sentences summarising the main findings. Discuss: did all three circuits work correctly? Were simulation and hardware results consistent? What was the dominant source of error for the monoflop timing? Comment on the role of the Schmitt-trigger at the clock input and the importance of the POR circuit.]

7 Summary

[3–5 sentence recap: restate the objective, briefly name the methods used, state the key results, and give the main conclusion. No new information may appear here.]

8 Outlook

[Suggest 2–3 extensions, for example: bidirectional chaser light using a multiplexer to select shift direction; catch counter with 7-segment display; variable catch time via potentiometer on the monoflop RC network.]

References

- [1] T. Floyd, *Digital Fundamentals*, 9th ed. Pearson Prentice Hall, 2006, ch. 7.
- [2] U. Tietze and C. Schenk, *Halbleiterschaltungstechnik*, 11. Auflage. Springer, 1999.
- [3] C. Thorrold, *Guide for Writing Lab Reports*, Version 1.0.
- [4] Texas Instruments, *74HC164 8-Bit Serial-In, Parallel-Out Shift Register – Datasheet*, <https://www.ti.com/lit/ds/symlink/sn74hc164.pdf>, accessed May 2026.
- [5] Texas Instruments, *74HC74 Dual D-Type Positive-Edge-Triggered Flip-Flop – Datasheet*, <https://www.ti.com/lit/ds/symlink/sn74hc74.pdf>, accessed May 2026.
- [6] Texas Instruments, *74HC221 Dual Monostable Multivibrator – Datasheet*, <https://www.ti.com/lit/ds/symlink/cd74hc221.pdf>, accessed May 2026.
- [7] Texas Instruments, *74HC160 Synchronous 4-Bit Decade Counter – Datasheet*, <https://www.ti.com/lit/ds/symlink/sn74hc160.pdf>, accessed May 2026.

A Appendix A – Schematics

A.1 Task 3.1 – Chaser Light

Figure 7: Complete schematic – chaser light circuit (Task 3.1).

A.2 Task 3.2 – Catch Circuit

Figure 8: Complete schematic – catch circuit (Task 3.2).

A.3 Task 3.3 – Catch Counter

Figure 9: Complete schematic – catch counter circuit (Task 3.3).

B Appendix B – Actual Component Values

Table 4: Component values measured with multimeter before assembly.

Component	Nominal	Measured	Tolerance	Task
R_{POR}	1 M Ω	?? M Ω	$\pm 5\%$	3.1–3.3
C_{POR}	1 μ F	?? μ F	$\pm 20\%$	3.1–3.3
R_{mono}	2 M Ω	?? M Ω	$\pm 5\%$	3.2–3.3
C_{mono}	1 μ F	?? μ F	$\pm 20\%$	3.2–3.3
R_{pull} (button)	10 k Ω	?? k Ω	$\pm 5\%$	3.2–3.3